



NETSEC HUB

Professional Networking & Cybersecurity Training

Cisco Certified Network Associate (CCNA)- Industry Networking Fundamentals

- ***Mode: Online***
- ***Duration: 2-2.5 Months***

Course Overview:

CCNA Industry Skill Development Program

This course is designed to build strong practical networking fundamentals required for entry-level Network Engineer roles. The focus of this program is not limited to certification exam preparation; instead, it focuses on understanding real-world networking concepts, device configuration, troubleshooting approaches, and enterprise network implementation.

Students will learn how networks are designed, configured, and maintained using industry-standard concepts followed in enterprise environments. The course includes live demonstrations, practical configuration sessions, troubleshooting scenarios, Q&A discussions, and interview preparation to help students develop job-oriented networking skills.

Hands-on concepts will be demonstrated using network simulation environments such as EVE-NG. Students will learn practical implementation of networking technologies through configuration demonstrations and troubleshooting scenarios based on the course topics. *(Lab files and personal lab infrastructure are not included with the course.)*

Course Modules:

Module 1: Networking Fundamentals

- Introduction to Computer Networks
 - Types of Networks (LAN, WAN, MAN)
 - Network Devices: Hub, Switch, Router, Firewall etc.
 - OSI Model and TCP/IP Model
 - IP Addressing Fundamentals
 - IPv4 Address Classes and Private/Public IP Addresses
 - Subnet Mask and Subnetting
 - Broadcast, Unicast and Multicast Communication
 - MAC Address and ARP Protocol
 - Basic Cisco IOS Navigation and Configuration
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Module 2: Network Protocols and Services

- DHCP Protocol and DORA Process, DHCP Relay Configuration
 - DNS Fundamentals
 - FTP and TFTP
 - SSH and Telnet
 - NTP
 - Basic Server Connectivity
 - Troubleshooting Network Services
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Module 3: Switching Fundamentals

- Introduction to Layer 2 Switching
- VLAN Concepts
- Access Ports and Trunk Ports
- Native VLAN
- Inter-VLAN Routing:
 - Router-on-a-Stick
 - Layer 3 Switch SVI
- VLAN Troubleshooting
- Spanning Tree Protocol (STP)
- Root Bridge Election, Port Roles and States
- EtherChannel:
 - Static EtherChannel
 - PAGP
 - LACP

Module 4: Routing Fundamentals

- Routing Concepts
 - Router Decision Making
 - Routing Table Understanding
 - Static Routing
 - Default Routing
 - Dynamic Routing Introduction
 - RIP Fundamentals
 - OSPF Fundamentals:
 - Neighbor Formation
 - OSPF States
 - Areas Concept
 - DR/BDR Election
 - EIGRP Fundamentals
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Module 5: Network Security Fundamentals

- Access Control Lists (ACL)
 - Standard ACL
 - Extended ACL
- Network Address Translation (NAT)
 - Static NAT
 - Dynamic NAT
 - PAT
- Basic Device Security
- Password Security
- SSH Configuration

Career Outcome

After completing this program, students will develop a strong foundation in enterprise networking concepts along with practical configuration and troubleshooting skills required for entry-level networking roles such as:

- Network Support Engineer
 - Network Engineer (L1)
 - IT Infrastructure Engineer
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